

Md. Mahiul Kabir

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github.com/rbwkai | linkedin.com/in/mahiulkabir | codeforces.com/rbwkai

Competitive programmer (Codeforces Expert) building rigorous C++ systems, full-stack apps, and embedded tools.

Education

Islamic University of Technology (IUT)

B.Sc. in Computer Science & Engineering — CGPA: 3.63/4.00

Expected 2027

Dhaka, Bangladesh

St. Joseph Higher Secondary School

Higher Secondary Certificate (HSC) — GPA: 5.00/5.00

2022

Dhaka, Bangladesh

Competitive Programming & Achievements

- Competed onsite in **14 collegiate contests** — 2 ICPC Asia Regionals, NCPC, and **11 IUPCs** — best ranks **17th** (DUET 2026), **18th** (NSU 2026), **19th** (SUST 2026, AUST 2025)
- Divisional Champion** (2018) and **3× National Qualifier** (2017, 2018, 2020), Bangladesh Mathematical Olympiad (BDMO)
- Codeforces Expert** (peak rating 1692, handle `rbwkai`) — 1,500+ problems solved across Codeforces, CSES, and other judges (230+ on CSES alone)

Experience

IUTPC (IUT Programming Club) — Instructor

Official competitive programming club of IUT CSE

Since Jun 2024

Dhaka, Bangladesh

- Instruct 50+ freshers across two batches in competitive programming fundamentals (dynamic programming, graph theory, data structures); several students have advanced to national-level onsite contests
- Author problem sets as Problemsetter for the IUT Intra Junior Programming Contest (2025, 2026) and serve as Judge for the IUT Intra Programming Contest (2025)

Project Altair — Software Sub-team Member

IUT's Mars Rover Team

Since 2024

Dhaka, Bangladesh

- Diagnosed and resolved critical bugs in the rover's ROS-based automation pipeline, improving subsystem control, coordination, and overall runtime reliability
- Built OS-level integration and embedded automation for the rover's onboard control system

Projects

Sada | C++17, CMake, SQLite | [GitHub](#)

- Designed **SAFS**, a novel spaced-repetition algorithm for an ungraded Arabic vocabulary trainer that fuses power-law forgetting curves, frequency-derived difficulty, and desirable-difficulty stabilization to schedule review without any self-reported grading — shipped with a bundled, corpus-grounded 2,000-word frequency-ranked MSA dataset

xLog | C++20, CMake, SQLite | [GitHub](#)

- Architected a terminal-native productivity engine that gamifies task management via dynamic difficulty calibration (CR-EMA signal model), domain-balanced XP, and grace-token streak resilience across four task archetypes, with a strict sub-300-line modular architecture and time-budget-aware morning-block scheduling

Girih | C++, OOP | [GitHub](#)

- Engineered a declarative C++ geometry engine that composes points, lines, and circles as OOP primitives, drives them with parametric `Value` objects, and renders keyframed procedural math animations in real time at 60 FPS, HD resolution

Pookieping | MERN Stack, Security | [GitHub](#)

- Built a deliberately vulnerable MERN social wall (JWT auth) as an XSS training ground, with a toggleable sanitization layer that switches between vulnerable and hardened rendering to demonstrate stored-XSS exploitation and defense

Technical Skills

Languages: C++ (11–20), Python, Java, Rust

Web & Backend: Spring Boot, React, React Native, Node.js/Express, MERN Stack

Databases: PostgreSQL, SQLite, SQL

Systems & Tools: Linux, Bash, Git, CMake, ROS, Raspberry Pi, CTF / Web Security Fundamentals

Beyond the Technical

Multidisciplinary artist — sketching, digital illustration, 3D art (Blender), origami; learning Arabic; speedcuber (14.41s WCA personal best); advanced calisthenics (full planche, front lever) and combat athlete (MMA, Taekwondo).